

BINAY - Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines). BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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two-digit counter-cum-display circuit is very similar to the seconds counter circuit. It comprises flip-flop CD4027 (IC13), two counter-cum-7-segment driver CD4033 ICs (IC14 and IC15) and two 7-segment displays (DIS3 and DIS4).

The circuit receives a pulse every twelve hours as clock input from the hours unit. The input is divided

by '2' (by flip-flop CD4027 (IC13)) to generate one pulse per day (PPD) at pin 1 of IC13. This PPD output is fed to the two-digit date counter (IC14). IC14 and IC15 count the one pulse per day to display the date. The AM or PM indication (using decimal point of DIS3) is provided by pin 2 of IC13. You can reset the date counter by pressing switch S5 momentarily and set the required date by connecting SPDT switch S8 to switch S11 and press switch S11 until you arrive at the desired date.

Day-of-the-week display. The pulse per 12 hours from the hours counter is inverted by gate N11 and then gate N12. Toggle switch S14 connected across inverting gate N12 is used for synchronisation of day with date by opening or closing it. The pulse per 12 hours is divided by flip-flop CD4027 to generate a PPD clock at its pin 15. This PPD is used as the input for decade counter CD4017 (IC16), which counts and lights up one LED for each day of the week. (Q0 indicates Sunday, Q1 through Q6 indicate Monday through Saturday and Q7 resets the day counter on every seventh input of one pulse per 24 hours.)

The day counter can also be reset manually by momentarily pressing switch S4. Set the day by connecting SPDT switch S9 to switch S10 and press switch S10 until the desired day LED glows. Move switch S9 to normal position after the setting is done.

Month and year display. The month-and-year display circuit is very similar to the date display circuit, except for IC13. The 1Hz input is fed to pin 1 of IC14 through a push-to-on switch to set the desired month (1-12) or year (00-99) manually after resetting the counter to '00.'

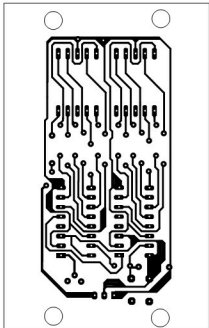


Fig. 7: Actual-size, single-side PCB for the circuit around IC14 and IC15